

Zhu Zhanyan

Software Engineer

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Results-driven Software Engineer with a strong foundation in Data Engineering and Artificial Intelligence applications.

EXPERIENCE

- Inland Revenue Authority of Singapore (IRAS)**

Data Engineer Intern

Jun 2025 - Jan 2026

- Migrated legacy SAS refund-risk processing logic to SQL, modernising data workflows and improving maintainability of production pipelines.
- Built a Jinja-based SQL transpiler enabling shared processing logic to execute across Spark and SQLite for distributed training and local inference workloads.
- Architected a Kedro pipeline integrating Azure ML and Azure Synapse to construct and refresh a LanceDB vector database for an internal Retrieval-Augmented Generation (RAG) application.
- Automated vector database refreshes with scheduled pipelines, eliminating manual operations and enabling reliable, repeatable data updates.

- Gojek**

Machine Learning Engineer Intern

Mar 2020 - Oct 2020

- Engineered an open-source Machine Learning (ML) Feature Store for Gojek's driver-customer matchmaking model, enhancing data processing efficiency using Python, Java, Golang, Kafka, and Redis.
- Resolved critical production issue by correcting Redis storage schema mismatch, ensuring uninterrupted matchmaking model performance.
- Designed and deployed a comprehensive analytics dashboard using Grafana and GCP BigQuery, empowering management with real-time insights for data-driven decision-making.

PROJECTS

- Broche: AI-Assisted Drawing App**

Aug 2025- Apr 2026

<https://www.mrzzy.co/work/augmenting-rather-replacing-artists-gen-ai>

<https://github.com/mrzzy/ntu-fyp>

- Engineered an iPad AI-assisted drawing app that allows artists to co-create with AI image generation.
- Developed a tool-calling Sketch Agent that autonomously calls AI rendering, editing features.
- Optimised UI rendering with caching in hot path, reducing and drawing rendering time by 4.58x.

- B+Tree Database in C++**

Feb 2025- Mar 2025

<https://www.mrzzy.co/work/implementing-btree-database-in-cpp>

<https://github.com/mrzzy/ntu-sc3020/commits/main/project1>

- Built a disk-backed B+Tree database in C++ with columnar storage and filesystem block-aligned I/O.
- Implemented bulk-loaded B+Tree indexing to support efficient range-based aggregation queries.
- Achieved 7x faster queries and 87% fewer data block reads using index scan vs. brute force scan.

EDUCATION

- Nanyang Technological University**

Bachelor of Computing in Computer Science

Aug 2023 - Dec 2026

- Ngee Ann Polytechnic**

Diploma in Information Technology

Apr 2018 - Apr 2021

AWARDS

- **ISC26 Student Cluster Competition:** Team placed joint 1st in international HPC Competition. Worked on HPL, HPCG and power management.

SKILLS

- **Backend:** Python, SQL, Java, JavaScript, TypeScript, Node.js, Golang, Django, Jinja, Rust
- **Cloud:** GCP, Firebase, AWS, Azure, Terraform, Docker, Ansible, Linux, Kubernetes, CI/CD
- **Data Engineering:** PostgreSQL, Redis, BigQuery, Grafana, DBT, Prefect, Spark, DuckDB, Firestore, Kedro, Data Modeling, Data Warehousing, ELT Pipelines
- **AI Engineering:** LangChain, LanceDB, AI Agent, RAG Pipelines, LLM, VLM, Diffusion